INCH-POUND

MIL-G-70617A (AR)
26 September 1991
SUPERSEDING
(See 6.7)

MILITARY SPECIFICATION

GUNFIRE SIMULATOR: 12725085

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for the Gunfire Simulator (GUFS), also known as Target Kill Simulator (TKS) or Armor Target Kill Simulator (ATKS), used as a component of the Remote Target System (RETS) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-70618	-	Circuit Card Assembly, Gunfire Simulator: 12725086

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1290

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-882	-	System Safety Program Requirements
MIL-STD-2000	-	Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.3)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12725085	-	Gunfire Simulator
12725014	-	Electronic Control Assembly
9342148	-	Igniter, Electrical Assembly

SPECIAL PACKAGING INSTRUCTIONS

PS 12725085	-	Special Packaging Instructions for the Gunfire Simulator

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General description. The gunfire simulator is an electronic device used to activate the M21 and smoke pyrotechnic cartridges to simulate the effects of gunfire.

3.2 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.3 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12725085, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the GUFS shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.4 Performance.

3.4.1 Cartridge (M21 Type) interfaces.

3.4.1.1 Electrical output interface. The GUFS shall provide an output pulse of 1.7 to 2.2 amperes for a duration of 110 to 140 milliseconds at each output in response to a valid firing command per 3.4.3.4.

3.4.1.2 Mechanical interface. The GUFS shall be capable of housing 20 M21-type cartridges or equivalent envelope units in its firing tubes. Output ports shall interface with all fielded M21 electrical connectors in accordance with drawing 9342148.

3.4.1.3 Simulated match characteristics. In lieu of using actual M21 cartridges for contractor testing, a simulated cartridge can be installed into each output port. Figure (1) schematic shall be used as the basis for fabrication of the simulated cartridge match. The simulated cartridge should be interconnected to the GUFS output ports by means of a connector conforming to the plug on drawing 9342148.

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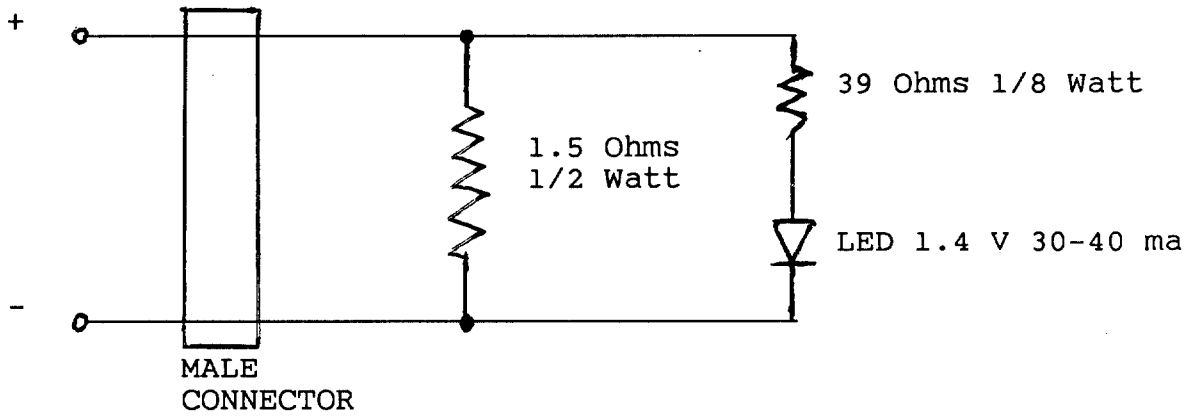


Figure 1. Simulated match schematic.

3.4.2 Electrical interface.

3.4.2.1 Input cable. The GUFS shall include an interface cable (minimum length ten meters) compatible with target mechanisms per 3.4.2.2 and compatible with GUFS output receptacle for use in tandem operation.

3.4.2.2 Electrical input interface. Electrical design of the GUFS shall be compatible with, and operate safely with, the Target Holding Mechanism/Tank Gunnery (THM/TG), the Armored Moving Target Carrier (AMTC) and the Improved Lifting Target Elevating Mechanism (ILTEM), per Table I. Input power from control equipment should be nominal 12 VDC ± 1.5 volts.

TABLE I. GUFS pin assignment.

The following connector pins have signals as defined below:

Pin	
A	Ground
B	Not Used
C	Not Used
D	Not Used
E	Not Used
F	Not Used
G	Not Used
H	Not Used
I	Smoke Fire Command
J	Smoke Fire Command
K	+12 VDC
L	Not Used
M	Internal input to relay; must be jumpered to Pin K
N	M21 Fire Command

NOTE: Pins identified as "Not Used" may have other applications with other devices. No connection is to be made to pins so designated.

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3.4.3 Functional control.

3.4.3.1 Electronic control. The GUFS shall be controlled by the Circuit Card Assembly (CCA). Electrical performance of the GUFS shall be in accordance with the performance characteristics specified in paragraph 3.3 of MIL-C-70618.

3.4.3.2 Firing sequence. Only a single port shall be activated each time a valid firing command is processed. All ports shall be activated in sequence for the selected operating mode, and no port shall be commanded more than once during any operating mode until the GUFS control system is reset by activation of the ARM/SAFE switch. Last port fired in sequence shall be retained by the GUFS memory circuit even with 12 volt system power disconnected, provided the ARM/SAFE switch is not moved from the ARM position. Operation of the ARM/SAFE switch shall reset the firing sequence to port 1.

3.4.3.3 Firing modes. Two firing modes shall be provided:

a. Sequential firing of ports 1 through 20.

b. Partitioned firing of ports 1 through 15, and ports 16 through 20.

3.4.3.3.1 Sequential mode. Firing signals on the input connector Pin N shall automatically fire in sequence each GUFS port once from 1 through 20. For series operation the signal shall be transferred from Pin N (J4) to Pin N (J3) of the tandem GUFS unit.

3.4.3.3.2 Partitioned mode. GUFS shall automatically partition into subsets of 1-15 and 16-20 upon receipt of a fire signal on Pin I or Pin J of the input connector. After partitioning;

a. Firing signals on input connector Pin N will fire ports 1-15 in sequence and then be transferred to Pin N of the input transfer connector for series operation when required.

b. Firing signals on input connector Pins I or J shall fire ports 16-20 in sequence and then be transferred to Pin I of the input-transfer connector for series operation when required.

3.4.3.4 Firing command. The input firing command shall have the following characteristics.

a.	Minimum voltage:	9.3 VDC
b.	Maximum voltage:	13.6 VDC
c.	Minimum pulse width:	50 milliseconds
d.	Maximum duration:	infinite

3.4.3.5 Output firing pulse. The output pulse shall be current limited to provide current of 1.7 to 2.2 amperes for 110 to 140 milliseconds to the commanded port to be fired.

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3.4.4 Safety features. GUFS shall conform to the safety requirements of MIL-STD-882 and incorporate positive safety features in accordance with the following:

a. SAFE/ARM switch shall be provided on electronic control assembly 12725014 which prevents firing signals from activating cartridges when switch is in the SAFE position.

b. Protection against damage or misfire when output connectors are directly shorted to frame or 12 VDC signal/power is interrupted and returns.

c. Maintain a short circuit to ground across each output port until such time as it is designated to fire. Power shall only be applied during the firing pulse cycle and only to the commanded port to be fired.

3.4.5 Steady-state current drain. GUFS shall have an overall steady-state current drain of not greater than seventy five (75) milliamperes. In addition, the maximum instantaneous current draw during any firing sequence shall not exceed 2.3 amperes.

3.5 Environmental requirements.

3.5.1 Temperature. The GUFS shall meet all requirements of 3.4 when thermally stabilized at -25°F and 149°F and when returned to and thermally stabilized at ambient temperature.

3.5.2 Solar radiation. The GUFS shall meet all requirements of 3.4 when and after being subjected to three 24 hour cycles of solar radiation, 1120 Watts per meter squared at 149°F, in accordance with MIL-STD-810, Method 505, Procedure II.

3.5.3 Vibration (loose cargo transport). The GUFS shall meet all the requirements of 3.4 after exposure to a loose cargo transport test as specified in MIL-STD-810, Method 514, test Procedure II except that it shall be vibrated with one inch double amplitude at 5 Hz in the Horizontal and transverse axis for a test duration of 90 minutes per axis.

3.5.4 Operational shock. The GUFS shall meet the requirements of 3.4 after being subjected to five simultaneous M21 charge firings when connected to a THM/TG or ILTEM assembly.

3.5.5 Sand and dust. The GUFS shall meet all the requirements of 3.4 after exposure to blowing dust as specified in Procedure I, Method 510 of MIL-STD-810: test temperature 145°F, air velocity per paragraph I-3.2c (high end of range), dust composition per paragraph I-3.2d, dust concentration per paragraph I-3.2e, and test duration per paragraph I-3.2h of Method 510. The test item configuration shall be as normally deployed (not protected).

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3.5.6 Immersion. The GUFS electronic control assembly, Part No. 12725014, shall meet the requirements of 3.4 after being submerged in water to the depth of 2 ft. of water for 2 hours at ambient temperature per MIL-STD-810, Method 512, Procedure I.

3.5.7 Leakage. The GUFS electronic control assembly shall show no evidence of leakage after being subjected to 1 to 2 psi of dry nitrogen for 24 hours.

3.5.8 Electrostatic discharge. The GUFS shall meet the requirements of 3.4 when subjected to 15 KV on a 100 ± 5 pfd capacitor through a 1500 ± 75 ohm resistor.

3.5.9 Electromagnetic emission and susceptibility. The GUFS unit shall meet the requirements of 3.4 when subjected to REO2, CEO3 and RSO3 of MIL-STD-461. Part IV Army Values for the electrical field shall apply.

3.6 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (Task) F paragraph 1.2 of MIL-STD-2000.

3.8 Workmanship. The workmanship requirements of MIL-F-13926, and applicable requirements for MIL-STD-454 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5) (if required).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (+16°C to +32°C)
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg (700 to 815 mmHg)

4.3 First article inspection.

4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
GUFS	12725085	3

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in Table II and IV of this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to

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product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawing and specification utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests (100% Inspection)

Characteristic		Requirement	Test Procedure
Critical:		None	N/A
Major:			
101.	Electronic Control	3.4.3.1	4.7
102.	Firing sequence	3.4.3.2	4.7
103.	Firing mode	3.4.3.3	4.7
104.	Firing commands	3.4.3.4	4.7
105.	Firing pulse output	3.4.3.5	4.7
106.	Steady-state current drain	3.4.5	4.7
107.	Safety features	3.4.4	4.7
108.	Leakage	3.5.7	4.8.9
109.	Identification plate	3.3	4.7.1
Minor:			
201.	Workmanship	3.8	Visual
202.	Materials	3.2	Visual

4.5 Special sampling. Three (3) GUFS out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE III. Special sampling.

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
109.	Immersion	3.5.6	4.8.8
110.	Electrostatic discharge	3.5.8	4.8.10
Minor:		None	N/A

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Table IV. Additional first article tests.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
111. High Temperature	3.5.1	4.8.2
112. Low Temperature	3.5.1	4.8.3
113. Solar Radiation	3.5.2	4.8.4
114. Vibration	3.5.3	4.8.5
115. Operational Shock	3.5.4	4.8.6
116. Sand and Dust	3.5.5	4.8.7
117. Electromagnetic emissions and susceptibility	3.5.9	4.8.11
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment

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shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.7 Methods of inspection.

4.7.1 GUFS functional test. Each GUFS unit shall be functionally tested using a contractor-provided and Government approved test fixture which shall be capable of providing the appropriate power and firing commands as well as monitoring the GUFS individual output port firing pulses using simulated M21 cartridge per 3.4.1.3 and transfer output at the output connector. A test cycle shall consist of the sequence outlined in Table V. Observation shall be made to determine compliance with the listed responses.

Sequential Firing Mode	- 40 Firings, Pin N
Partitioned Firing Mode	- 30 Firings, Pin N
Partitioned Firing Mode	- 10 Firings, Pin I/J

At the contractor's option, with Government approval, two (2) GUFS may be functionally tested simultaneously in a tandem master slave configuration provided the master slave orientation is reversed during 50% of the test cycle. Prior to performing the test, verify that the name plates for the Simulator Gunfire (GUFS) and the Electronic control Assy 12725014 have been stamped correctly and completely for each unit. During the test sequence, the following characteristics will be checked either manually or automatically by an approved test fixture for conformance with specified requirements and allowable limits.

Functional Characteristic	Requirement
a. Electronic Control	3.4.3.1
b. Firing Sequence	3.4.3.2
c. Firing Mode	3.4.3.3
d. Firing Command	3.4.3.4
e. Firing Pulse Output	3.4.3.5
f. Safety Features	3.4.4
g. Steady-state Current Drain	3.4.5

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TABLE V. Functional Test Procedure/Data Sheet - GUFS.

Serial No._____

Test Condition:_____ Date Tested:_____

Test Location:_____ Inspector:_____

Sequence	Operation	Response	Verified
1.	Turn simulator power On.	POWER - ON	
2.	With Arm/Safe Switch in safe position, Fire M21.	NO FIRE	
3.	Fire smoke	NO FIRE	
4.	Arm GUFS and fire M21 (T < 50 SEC)	NO FIRE	
5.	Fire M21 (T < 50 SEC)	NO FIRE	
6.	Wait 75 seconds and Fire M21.	FIRE - ON 1	
7.	Fire M21 twice rapidly	FIRE - ON 2	
		NO FIRE 3	
8.	Fire M21	FIRE - ON 3	
9.	Fire smoke	FIRE - ON 16	
10.	Fire 3 M21	FIRE - ON 4	
		FIRE - ON 5	
		FIRE - ON 6	
11.	Fire smoke	FIRE - ON 17	
12.	Fire 3 M21	FIRE - ON 7	
		FIRE - ON 8	
		FIRE - ON 9	
13.	Fire smoke	FIRE - ON 18	
14.	Turn power off	POWER - OFF	
15.	Turn power on	POWER - ON	
16.	Wait 75 seconds and fire 3 M21	FIRE - ON 10	
		FIRE - ON 11	
		FIRE - ON 12	
17.	Fire smoke	FIRE - ON 19	
18.	Fire 3 M21	FIRE - ON 13	
		FIRE - ON 14	
		FIRE - ON 15	
19.	Fire smoke	FIRE - ON 20	
20.	Fire 1 M21	FIRE-ON M21 Transfer	
21.	Fire 1 smoke	FIRE-ON Smoke Transfer	
22.	Place ARM/SAFE switch to safe	--	
23.	Turn Power Off - wait 15 seconds	POWER - OFF	
24.	Place ARM/SAFE switch to arm		
25.	Turn power on and wait 75 seconds	POWER - ON	
26.	Fire 1 M21	FIRE - ON 1	
27.	Fire 1 smoke	FIRE - ON 16	

NOTE: Except for step 7, minimum of 2 seconds allowed between each firing pulse.

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4.8 Environmental.

4.8.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.8.2 High temperature. Per 3.5.1, the device shall be tested in accordance with MIL-STD-810, Method 501, Procedure I, with a maximum operating temperature of 65°C (149°F). While operating at the maximum temperature, the device shall successfully complete one cycle of the functional test as specified in 4.7. Following completion of the high temperature test, and upon return to ambient conditions, the device shall be tested as specified in 4.7 to determine compliance with 3.5.1.

4.8.3 Low temperature. Per 3.5.1, the device shall be tested in accordance with MIL-STD-810, Method 502, Procedure I, with a minimum operating temperature of -32°C (-25°F). While operating at the minimum temperatures, the device shall successfully complete one cycle of the functional test specified in 4.7. Following completion of the low temperature test, and upon return to ambient conditions, the device shall be functionally tested as specified in 4.7 to determine compliance with 3.5.1.

4.8.4 Solar radiation. Per 3.5.2, the device shall be tested at 65°C (149°F) for three cycles of solar radiation specified in accordance with MIL-STD-810, Method 505, Procedure II. This test shall be conducted on the First Article only. During the maximum temperature portion of each of the 24-hour cycles, and upon return to ambient conditions, the device shall be functionally tested as specified in 4.7 herein to determine compliance with 3.5.2.

4.8.5 Vibration (loose cargo transport). This test shall be performed as specified in Procedure II, Method 514 of MIL-STD-810 except that the GUFS shall be vibrated in the horizontal and transverse axis with a one (1) inch double amplitude displacement at 5 Hertz for a test duration of 90 minutes in each axis. The orientation of the GUFS shall be in the normal upright position. Following completion of the vibration test the GUFS shall be functionally tested (one cycle) as specified in 4.7 herein to determine compliance with 3.5.3.

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4.8.6 Operational shock. A high level shock test shall be conducted by simultaneously firing five (5) live M21 cartridges randomly installed in the GUFS. The charges will be fired manually bypassing the normal GUFS control circuitry. After being subjected to the maximum shock pulse of five (5) simultaneous firings, there shall be no evidence of physical damage or electrical degradation of GUFS performance (such as cracked tubes, broken frame base or welds, or cracked electrical tails or components. Following completion of the operational shock test, the device shall be functionally tested as specified in 4.7 herein to determine compliance with 3.5.4.

4.8.7 Sand and Dust. This test shall be performed as specified in Procedure I, Method 510 of MIL-STD-810. Operation not required during test. Test temperature 145°F. Air velocity per Section I-3.2c (high end of range), dust composition per Section I-3.2d, sand and dust concentration per Section I-3.2e, test duration per Section I-3.2h of Method 510 of MIL-STD-810. The GUFS configuration shall be as normally deployed (not protected). Following completion of the dust test, and upon return to ambient conditions, the GUFS shall be functionally tested as specified in 4.7 herein to determine compliance with 3.5.5.

4.8.8 Immersion. Per 3.5.6, the device shall be tested or the immersions requirement specified in accordance with MIL-STD-810, Method 512, Procedure I. Upon completion of the test, the interior of the device shall be examined for evidence of water none is permitted. After successful completion of the immersion test, the test device shall be tested as specified in 4.7 herein to determine compliance with 3.5.6.

4.8.9 Leakage (pressure). To determine conformance to 3.5.7, on production units, each GUFS electronic control assembly shall be pressurized with 1 to 2 psi nitrogen. There shall be no measurable loss in differential pressure after 24 hours.

4.8.10 Electrostatic discharge. To determine conformance to 3.5.8 the device shall be tested by applying an electrostatic discharge (15 KV on a 100 ± 5 pfd capacitor through a 1500 ± 75 ohm non-inductive resistor) to the following areas:

a. Input connector.
b. Arming switch - Safe position.
c. Arming switch - Arm position.
d. Electronic Control Module Housing.
e. Frame Assembly.
f. Right Receptacle Plate Assembly.
g. Left Receptacle Plate Assembly.

After application of the electrostatic discharge to each of the above specified areas, the device shall be examined for damage and be tested as specified in 4.7 herein to determine compliance with 3.5.8.

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4.8.11 Electromagnetic emissions and susceptibility (EMI). To verify that the GUFS when operated is not a source of electro-magnetic interference, nor will the performance of the device be degraded by external electro-magnetic fields. The device shall meet the requirements of REO2, CEO3, and RSO3 of MIL-STD-461, Part 4, when subjected to Army values for the electric field. Testing will be conducted in accordance with the procedures outlined in MIL-STD-462, for U.S. Army Equipment when operating the GUFS in accordance with 4.7. GUFS shall comply with the following criteria, or be rejected.

a. REO2

Narrow band E-field emissions shall not be in excess of limit curve 4-12 of MIL-STD-461, Part 4.

Broadband E-field emissions shall not be in in excess of limit curve 4-13 of MIL-STD-461, Part 4.

b. CEO3

Narrow band and broadband emissions on DC and AC leads shall not be in excess of limit curve 4-3 and 4-4 of MIL-STD-461, Part 4.

c. RSO3

Device shall not exhibit malfunction or degradation of performance during operation, when subjected to radiated electric fields as follows:

Frequency	E-Field (Volt/Meter)
14 KHz-2 MHz	1
2-30 MHz	10
30-2000 MHZ	5
2-10 GHz	5
Above 10 GHz	20

After completion of test, the GUFS shall be functionally tested as specified in 4.7 herein to determine compliance with 3.5.9.

4.9 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in para 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 12725085. The level of protection shall be as specified in the procurement document.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Gunfire Simulator 12725085 provides simulated gunfire effects required during armored vehicle, infantry and helicopter training scenarios. It is compatible with the THM/TG, AMTC and ILTEM target mechanisms used as a part of RETS.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certification of conformance for each lot of shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Definitions.

6.5.1 Trainers. Training device terms are defined in MIL-HDBK-220. Other definitions, if required, will be furnished upon written request to the Procuring Contracting Officer.

6.5.2 Contractor. Unless otherwise specified, the word Contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government Owned-Contractor Operated (GOCO), and Government Owned-Government Operated (GOGO) plants.

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6.5.3 Technical data package. A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists, specifications, standards, models, performance requirements, quality assurance provisions and packaging data.

6.6 Subject term (key word) listing.

Armor Target Kill Simulator (ATKS)
Target Kill Simulator (TKS)
Remote Target System (RETS)
Target Holding Mechanism/Tank Gunnery (THM/TG)
Improved Lifting Target Elevating Mechanism (ILTEM)
Circuit Card Assembly (CCA)
Armored Moving Target Carrier (AMTC)

6.7 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1290-A452)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-G-70617A (AR)	2. DOCUMENT DATE (YYMMDD) 26 SEPTEMBER 1991

3. DOCUMENT TITLE
GUNFIRE SIMULATOR: 12725085

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US ARMY ARDEC STANDARDIZATION OFFICE	b. TELEPHONE (Include Area Code) (1) Commercial (201) 724-6674 (2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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